

Chia-Hsiang Kao

Ph.D. in Computer Science, Cornell University · Expected Fall 2027

ck696@cornell.edu
iandrover.github.io
Google Scholar · GitHub

RESEARCH INTERESTS

I build world models that infer and predict the hidden dynamics of real-world interactions: recovering 3D motion and physics from observation, and generating controllable video.

- **3D & physical world modeling:** completing occluded scene geometry and motion, and recovering rigid-body physics from video [CVPR 2026].
- **Video generation & in-context learning:** unifying video tasks through spatiotemporal analogy [under review].
- **LLM agents:** code-executing, tool-using agents and their evaluation [ACL 2026 Findings].
- **AI for science:** remote sensing [NeurIPS 2024 D&B], DNA modeling [ICML 2024, ICLR 2024 Workshop], and medical imaging [MICCAI 2021, EMBC 2022].

EDUCATION

Ph.D. in Computer Science | Cornell University, Ithaca, NY 2023 – Fall 2027 (expected)

Advisor: [Bharath Hariharan](#)

Doctor of Medicine (M.D.) | National Yang-Ming Chiao-Tung University, Taipei, Taiwan 2015 – 2022

RESEARCH EXPERIENCE

Student Researcher | Meta, New York, NY Jan 2026 – May 2027

Host: [Menglin Jia](#)

- **Amodal scene flow completion (ongoing):** leading scene-level completion of occluded geometry together with its 3D motion, beyond prior visible-only scene flow; designed a novel point-cloud tokenizer and a diffusion model that completes point-voxel tokens while predicting the motion field.
- **ViGeo (Preprint):** led a video in-context learning framework that unifies video tasks as spatiotemporal canvas completion; developed a single model that absorbs a diverse task taxonomy spanning camera control, modality-guided generation, and video manipulation, and generalizes to unseen tasks and sensing modalities; fully fine-tuned Wan2.2-5B on 32 A100 nodes.

Applied Scientist Intern | Amazon, Sunnyvale, CA May 2025 – Aug 2025

Host: [Chien-Yi Wang](#)

- **Dynamics (CVPR 2026):** led a VLM that recovers rigid-body physics from monocular video by generating MuJoCo scene configurations as structured text. Trained on 400K synthetic videos with natural-language motion reasoning as auxiliary supervision, it substantially outperforms frontier VLMs, improves further with test-time search, and transfers to a new benchmark of 235 real-world videos.

Graduate Researcher | Cornell University, Ithaca, NY 2023 – Present

Advisor: [Bharath Hariharan](#)

- **UnivEarth (ACL 2026 Findings):** led a benchmark of 140 Earth-observation questions across 17 satellite sensors for code-executing LLM agents using the Google Earth Engine API; showed frontier agents reach only 33% accuracy, with generated code failing 58% of the time.
- **AllClear (NeurIPS 2024 D&B):** led a dataset and benchmark for cloud removal in satellite imagery.
- **Counter-current learning (NeurIPS 2024):** proposed a biologically plausible alternative to backpropagation using anti-parallel feedforward and feedback networks.

MENTORING

- Hangyu Zhou: AllClear (NeurIPS 2024 Datasets & Benchmarks, co-first author)
- Cheryl Lam and Aarush Umap: UnivEarth (ACL 2026 Findings)

SKILLS

Modeling: world models, video generation (flow matching, Wan2.2), VLMs, LLM agents & tool use

Tools: PyTorch, FastVideo, multi-node distributed training (32 A100 nodes), MuJoCo, Google Earth Engine

PUBLICATIONS

[Google Scholar](#): 500+ citations, h-index 7 (Oct 2026)

2026 Unifying Video Tasks via Spatiotemporal Analogy

[Under review](#)

[Chia-Hsiang Kao](#), Belinda Zeng, Bharath Hariharan, Menglin Jia

Dynamics: Language-Based Representation for Inferring Rigid-Body Dynamics From Videos

[CVPR 2026](#)

[Chia-Hsiang Kao](#), Cong Phuoc Huynh, Chien-Yi Wang, Noranart Vesdapunt, Stefan Stojanov, Bharath Hariharan, Oleksandr Obiednikov, Ning Zhou

2025 Towards LLM Agents for Earth Observation

[ICML 2025 TerraBytes Workshop](#); [ACL 2026 Findings](#)

[Chia-Hsiang Kao](#), Wenting Zhao, Cheryl Lam, Aarush Umap, Shreelekha Revankar, Samuel Speas, Snehal Bhagat, Rajeev Datta, Cheng Perng Phoo, Utkarsh Mall, Carl Vondrick, Kavita Bala, Bharath Hariharan

2024 Counter-Current Learning: A Biologically Plausible Dual Network Approach for Deep Learning

[NeurIPS 2024](#)

[Chia-Hsiang Kao](#), Bharath Hariharan

AllClear: A Comprehensive Dataset and Benchmark for Cloud Removal in Satellite Imagery

[NeurIPS Datasets and Benchmarks Track 2024](#)

Hangyu Zhou*, [Chia-Hsiang Kao](#)*, Cheng Perng Phoo, Utkarsh Mall, Bharath Hariharan, Kavita Bala

Caduceus: Bi-Directional Equivariant Long-Range DNA Sequence Modeling

[ICML 2024](#)

Yair Schiff, [Chia-Hsiang Kao](#), Aaron Gokaslan, Tri Dao, Albert Gu, Volodymyr Kuleshov

2023 Advancing DNA Language Models: The Genomics Long-Range Benchmark

[AAAI 2023 Workshop](#); [ICLR 2024 MLGenX Workshop](#)

[Chia-Hsiang Kao](#)*, Evan Trop*, McKinley Polen*, Yair Schiff*, Bernardo P. de Almeida, Aaron Gokaslan, Thomas Pierrot, Volodymyr Kuleshov

FedBug: A Bottom-Up Gradual Unfreezing Framework for Federated Learning

[Preprint](#)

[Chia-Hsiang Kao](#), Yu-Chiang Frank Wang

2022 MEG-Based Classification and Grad-CAM Visualization for Major Depressive and Bipolar Disorders with Semi-CNN

[IEEE EMBC 2022](#)

Chun-Chih Huang, Intan Low, [Chia-Hsiang Kao](#), Chuan-Yu Yu, Tung-Ping Su, Jen-Chuen Hsieh, Yong-Sheng Chen, Li-Fen Chen

MAML Is a Noisy Contrastive Learner in the Classification

[ICLR 2022](#)

[Chia-Hsiang Kao](#), Wei-Chen Chiu, Pin-Yu Chen

2021 Demystifying T1-MRI to FDG18-PET Image Translation via Representational Similarity

[MICCAI 2021](#)

[Chia-Hsiang Kao](#), Yong-Sheng Chen, Li-Fen Chen, Wei-Chen Chiu

HONORS & AWARDS

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|--|------------|
| • Foundational AI PhD Fellowship, Cornell University | 2025 |
| • Student Travel Award, MICCAI | 2021 |
| • Undergraduate Research Fellowship, National Science and Technology Council, Taiwan | 2018, 2020 |
| • Summer Research Fellowship, National Health Research Institutes, Taiwan | 2018 |

SERVICE

Conference Reviewer: NeurIPS (2021, 2024–2026), ICLR (2025–2027), ICML (2025), ECCV (2026), COLM (2026), AAAI (2025), AISTATS (2025), AutoML (2022)

Journal Reviewer: CVIU (2022), Computers & Electrical Engineering (2024), IEEE TETCI (2024)